

Project Design Phase

Solution Architecture

Date	29 June 2026
Project Name	OptiCrop
Maximum Marks	4 Marks

Solution Architecture

The **Solution Architecture** for **OptiCrop – Smart Agricultural Production Optimization Engine** defines how different system components work together to provide intelligent crop recommendations. It connects agricultural challenges with modern technologies such as Machine Learning, cloud services, databases, and weather APIs to deliver accurate, secure, and scalable solutions for farmers.

- Develop an intelligent system to solve crop selection and resource management challenges.
- Analyze soil nutrients, pH, temperature, humidity, and rainfall to recommend suitable crops.
- Use Machine Learning to generate accurate, data-driven predictions.
- Provide a secure and user-friendly web/mobile application.
- Store agricultural and user data in a cloud database for efficient management.

OptiCrop – Solution Architecture Diagram

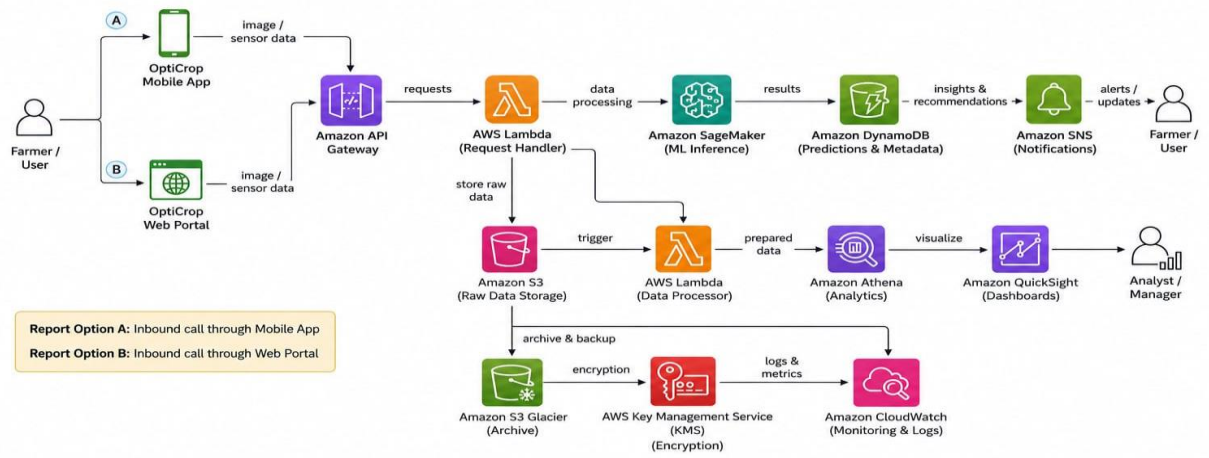


Figure 1: Architecture and data flow of the OptiCrop application